

NEURAL RESPONSES TO THE MECHANICAL PARAMETERS OF A HIGH-VELOCITY, LOW-AMPLITUDE SPINAL MANIPULATION: EFFECT OF PRELOAD PARAMETERS

William R. Reed, DC, PhD,^a Cynthia R. Long, PhD,^b Gregory N. Kawchuk, DC, PhD,^c and Joel G. Pickar, DC, PhD^d

ABSTRACT

Objective: The purpose of this study was to determine how the preload that precedes a high-velocity, low-amplitude spinal manipulation (HVLA-SM) affects muscle spindle input from lumbar paraspinal muscles both during and after the HVLA-SM.

Methods: Primary afferent activity from muscle spindles in lumbar paraspinal muscles were recorded from the L₆ dorsal root in anesthetized cats. High-velocity, low-amplitude spinal manipulation of the L₆ vertebra was preceded either by no preload or systematic changes in the preload magnitude, duration, and the presence or absence of a downward incisural point. Immediate effects of preload on muscle spindle responses to the HVLA-SM were determined by comparing mean instantaneous discharge frequencies (MIF) during the HVLA-SM's thrust phase with baseline. Longer lasting effects of preload on spindle responses to the HVLA-SM were determined by comparing MIF during slow ramp and hold movement of the L₆ vertebra before and after the HVLA-SM.

Results: The smaller compared with the larger preload magnitude and the longer compared with the shorter preload duration significantly increased ($P = .02$ and $P = .04$, respectively) muscle spindle responses during the HVLA-SM thrust. The absence of preload had the greatest effect on the change in MIF. Interactions between preload magnitude, duration, and downward incisural point often produced statistically significant but arguably physiologically modest changes in the passive signaling properties of the muscle spindle after the manipulation.

Conclusion: Because preload parameters in this animal model were shown to affect neural responses to an HVLA-SM, preload characteristics should be taken into consideration when judging this intervention's therapeutic benefit in both clinical efficacy studies and in clinical practice. (*J Manipulative Physiol Ther* 2014;37:68-78)

Key Indexing Terms: *Spinal Manipulation; Dosage Forms; Muscle Spindles; Neurophysiology; Spine; Cat; Chiropractic*

^a Associate Professor, Palmer Center for Chiropractic Research, Palmer College of Chiropractic, Davenport, Iowa.

^b Professor, Palmer Center for Chiropractic Research, Palmer College of Chiropractic, Davenport, Iowa.

^c Associate Professor, Faculty of Rehabilitation Medicine, University of Alberta, Edmonton, Alberta, Canada.

^d Professor Emeritus, Palmer Center for Chiropractic Research, Palmer College of Chiropractic, Davenport, Iowa.

Submit requests for reprints to: Joel G. Pickar, DC, PhD, Palmer Center for Chiropractic Research 741 Brady St, Davenport, IA 52803 (e-mail: jgpickar@live.com).

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Spinal manipulation (SM) is a form of manual therapy that patients often seek for musculoskeletal problems.¹⁻³ By its very nature, SM is a mechanical intervention. Delivered as a high-velocity, low-amplitude (HVLA) technique (HVLA-SM), it is used by most chiropractors⁴ as well as some osteopathic physicians and physical therapists.^{5,6} The preponderance of evidence indicates that HVLA-SM reduces pain and improves functional health status,⁷ but we know little of the mechanisms by which this occurs. Clinical improvements are thought to be initiated by the high-frequency peripheral sensory input evoked during the manipulation itself, improved spinal biomechanics, and sustained by a subsequent cascade of central neurophysiological changes.⁸⁻¹¹

Delivering HVLA-SM to a patient involves a number of mechanical considerations including the magnitude and duration of any preload that precedes the manipulation, the

HVLA-SM's application rate, amplitude and direction, the type of leverage used, and the contact site on both the clinician and patient.¹²⁻¹⁴ Analogous to the way in which a drug's chemical characteristics determine its pharmacokinetic and pharmacodynamic behavior, determining the way in which an HVLA-SM's mechanical characteristics influence neural and biomechanical responses should help identify the mechanical qualities that define HVLA-SM dosage. Currently, we know very little about the mechanical characteristics critical for the most effective application of SM. In clinical studies, descriptions of the HVLA-SM often lack adequate reporting for assessing how its mechanical characteristics relate to clinical outcomes.¹⁵

From neurophysiological studies using anesthetized animals where neural recordings from spinal tissues can be obtained directly, we know that as thrust durations approach those used clinically, the discharge frequency of muscle spindles in paraspinal muscles greatly increases.¹⁶⁻¹⁹ The increase depends more upon the amplitude of the applied thrust displacement than the applied thrust force.¹⁶⁻¹⁹ From biomechanical studies in humans, thrust force and duration appear to interact in changing spinal stiffness.²⁰ Increasing thrust force appears to increase intersegmental accelerations, whereas changing thrust duration has little effect on these accelerations.²¹ However, paraspinal electromyographic responses to an HVLA-SM are affected by both thrust force and duration.²¹ The relationship between any of these physiological responses and clinical outcomes is unknown.

In the current study, we were interested in how 1 mechanical consideration, the preload that often precedes delivery of an HVLA-SM, affects peripheral sensory input from lumbar paraspinal tissues both during and after the HVLA-SM. Preload characteristics of an HVLA-SM have been identified in several studies of chiropractic technique and appear to vary among chiropractors. Herzog's laboratory showed that preload forces range from 20 to 180 N, roughly comprising 9% to 32% of the thrust force^{22,23} and lasting between 0.5 and 5 seconds.²⁴ Gudavalli²⁵ showed slightly wider variation in preload forces ranging from 20 to 275 N but lasting 0.2 to 0.4 seconds. In addition to preload amplitude and duration, a third mechanical feature of the preload has been identified in force-time tracings. A brief partial unloading (previously referred to in the literature as "downward incisural point" [DIP]²⁶) may occur just before the thrust where a clinician rapidly reduces the preload just before applying the thrust.²⁶ Biomechanically, this decrease is thought to be undesirable.

We used the neural output from 1 type of proprioceptor, the muscle spindle, in dorsal back muscles to determine the way in which an HVLA-SM's preload characteristics can influence primary afferent input. Muscle spindles lie parallel to extrafusal fibers and respond to muscle length and the rate of change in muscle length. As a result, spindles potentially supply information to the central nervous system regarding joint position.^{27,28} Paraspinal muscle spindles

behave thixotropically,²⁹ acting stiffer and becoming more responsive when the parent muscle is maintained at an elongated position but becoming slack and less responsive when the muscle is shortened following an elongated history. In paraspinal muscles, changes in muscle spindle responsiveness occur following very small, sustained changes in vertebral position and are graded with the magnitude of change in vertebral position.³⁰ The change in responsiveness is also graded with the duration over which the change in vertebral position is maintained.³¹ The effect is maximal by approximately 4 seconds of lengthening history with a time constant of 1.1 second.³² Therefore, we hypothesized that muscle spindle discharge during an HVLA-SM would be greater as either preload magnitude or preload duration became greater. Because the preload DIP would be expected to create slack in the spindle immediately prior the manipulative thrust, we hypothesized that an HVLA-SM would be less effective at stimulating muscle spindles during an HVLA-SM when the preload contained a DIP than when it did not contain a DIP. Therefore, the purpose of this study was to determine how the preload that precedes an HVLA-SM affects muscle spindle input from lumbar paraspinal muscles both during and after the HVLA-SM.

METHODS

Electrophysiological responses from single muscle spindle afferents in lumbar paraspinal muscles were obtained before, during, and after a lumbar SM applied to 20 anesthetized (nembutal [35 mg/kg, intravenous {IV}]) cats of either sex weighing an average of 4.42 kg (SD, 0.72; range, 2.8-6.3 kg). Similar to previous studies,^{16-18,30,33} 1 afferent was investigated per cat because following completion of all SMs, surgery of the intact spinal tissues was needed to confirm that afferent activity was from muscle spindles in the lumbar multifidus or longissimus muscles. Cats were euthanized at the completion of each experiment with a bolus injection (0.5 mL, IV) of Sleepaway (Fort Dodge, IA) followed by saturated potassium chloride (KCl [3 M] IV) to fibrillate the heart. All experiments were approved by the Palmer's Institutional Animal Care and Use Committee (no. 20070101).

The experimental preparation including the basis for the surgical approach,^{34,35} the procedures used identifying primary afferent neurons as muscle spindles, and the equipment and methods used for applying an SM^{34,35} has been presented previously and is recently available through an open-access journal.¹⁶ Here, we present an overview of the approach, with detailed description of the choice of preload parameters, experimental design, and data analysis.

Muscle spindle activity from paraspinal muscles innervated by the L₆ spinal nerve was recorded in thin filaments of L₆ dorsal rootlets.³⁶ The laminectomy that exposed the L₆ rootlets kept the paraspinal muscles containing the muscle spindles intact because the L₆ dorsal root enters the

spinal cord 1 to 1½ vertebral segments cranial to the L₆ paraspinal soft tissues. Spindle responses represented their passive activity because surgical anesthesia and cutting the dorsal roots abolished reflex gamma responses.

An electronic feedback control system (Lever System Model 310; Aurora Scientific, Ontario, Canada) was used to apply the preload, HVLA-SM, and ramp and hold-type challenge movements to the L₆ vertebra, as previously described.^{16,33} The ramp and hold challenges were applied before and after the entire SM to determine whether the preload variables evoked sustained changes in muscle spindle signaling. All mechanical inputs were all applied in posterior to anterior direction to the skin overlying the L₆ spinous process.

Preload

Three preload variables were investigated: preload magnitude, preload duration, and presence or absence of a DIP. Based upon work from Herzog's laboratory where preloads in humans ranged from 9% to 32% of the thrust force^{22,23} and lasted between 0.5 and 5 s,²⁴ we used preload magnitudes of 18% and 43% of the peak applied thrust force (PATF) over preload durations of 1 and 4 seconds. The DIP removed 5% of the preload, which constituted 9% PATF. This was comparable with the 5% to 6% DIP reported in human studies.²⁶ The DIP reached its nadir in 20 milliseconds. Before the preload, we ensured that sufficient contact force (contact load) was applied so that any additional load or movement of the manipulator would move the vertebra, using a method as described previously.¹⁶

Spinal Manipulation

Force-time profiles measured in humans during HVLA-SM suggest that it is reasonable to apply an HVLA-SM as a triangular wave.^{14,23,37} With the motor in force control, it applied a linear increase in force reaching a peak applied force equaling 55% of a cat's average body weight (ABW). ABW (3.97 kg) was based on weights obtained from 112 cats in previous experiments.^{16,33} This relative force level is within the range used clinically after normalizing reported forces^{23,38} to a 70-kg person. Peak applied force (21.84 N) was reached in 75 milliseconds. The choice of this duration was based upon 2 considerations. In clinical studies using HVLA-SM, thrust durations in the thoracic and lumbar spines are typically less than 150 milliseconds.^{14,23,37} In animal studies,^{16,17} 75 milliseconds is a duration below the threshold duration at which an HVLA-SM induces a relatively high-frequency discharge in muscle spindles. Following peak of the thrust, the resolution force quickly returned to premanipulation levels within 25 milliseconds.

Ramp and Hold

To determine the effect of SM on muscle spindle signaling, identical dynamic and static mechanical stimuli

Table 1. *Protocols used for each experiment*

Protocol no.	Preload characteristic		DIP	HVLA-SM
	Preload magnitude (% PATF)	Preload duration (s)		
1 (control)	0	0	No	No
2 (control)	0	0	No	Yes
3	18	1	Yes	Yes
4	18	1	No	Yes
5	18	4	Yes	Yes
6	18	4	No	Yes
7	43	1	Yes	Yes
8	43	1	No	Yes
9	43	4	Yes	Yes
10	43	4	No	Yes

Protocols 3 to 10 were presented in random order.

Abbreviations: DIP, downward incisural point; HVLA-SM, high-velocity, low-amplitude spinal manipulation; PATF, peak applied thrust force.

were applied to the L₆ muscle spindle before and after the manipulation. The mechanical stimulus consisted of ramp and hold movement applied to the L₆ vertebra. Before the challenge, we placed a contact load on the vertebra and with the motor in displacement control, slowly (0.5 mm/s) moved the motor's lever arm ventralward 2.0 mm, and held the vertebra at its new position for 4 seconds. The lever arm was then slowly returned to its original position. The first ramp and hold preceded the SM by 5 minutes. The second ramp and hold was applied 10 seconds following the manipulation. This represented the minimal duration required to convert the motor from force control used for SM to displacement control used for the ramp and hold as well as to re-establish contact load.

Study Protocols. Each cat received 10 protocols consisting of ramp and hold testing, combinations of preload variables, and HVLA-SM (Table 1). The first protocol always represented a time control consisting, therefore, of ramp and hold testing but no HVLA-SM nor preload. The second protocol always represented a preload control consisting, therefore, of ramp and hold and HVLA-SM but no preload. Their presentation was not randomized as part of the experimental design. The next 8 protocols consisted of ramp and hold testing, HVLA-SM, and 8 combinations of the 3 preload variables (2 levels of magnitude, 18% and 43% peak force; 2 levels of duration, 1 and 4 seconds; and presence or absence of DIP). The 8 different combinations preload variables were presented in random order across cats.

Data Analysis. Spindle discharge was first quantified as instantaneous frequency (IF) by taking the reciprocal of the time interval between successive action potentials. To determine the effect of preload on muscle spindle responses during the 75-millisecond manipulative thrust, average afferent activity (mean instantaneous discharge frequencies [MIF]) was quantified during the 2 seconds preceding each manipulation (baseline) and during the manipulation's thrust phase. Only the thrust phase was used because the spindles became silent during the manipulation's resolution

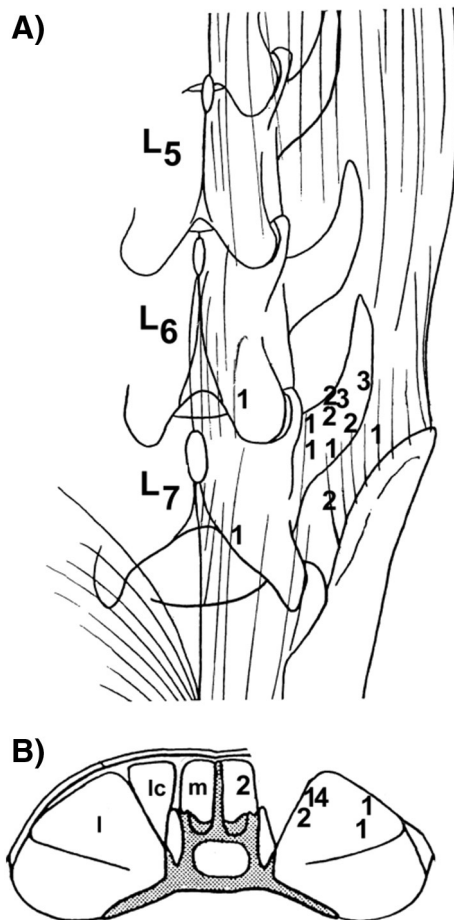


Fig 1. Location of the most sensitive portion of the receptive fields of the 20 lumbar paraspinal muscle spindle afferents. Numbers represent the number of afferents with their receptive field at that location. A, Dorsal view. B, Cross-sectional view roughly aligned with dorsal view. m, multifidus; l, longissimus; lc, lumbococcygeus that was removed when the receptive fields were located.

phase. Muscle spindle afferents can have a brief burst of activity at the start of movement associated with the movement's acceleration.^{39,40} We were interested in the spindle's response to the constant velocity portion of the thrust and, therefore, calculations of MIF during the thrust excluded its first 12.5 milliseconds. The change in MIF during the manipulation ($\Delta\text{MIF}_{\text{during}}$) served as the response measure and was calculated as the difference between the thrust phase and baseline.

To determine the effect of an HVLA-SM's preload on static and dynamic aspects of spindle responses during vertebral movement, MIF was determined during specific portions of each ramp and hold. Static aspects were represented by resting spindle discharge measured before the onset of the ramp ($\text{MIF}_{\text{resting}}$) and by spindle discharge measured during the hold when the vertebra was maintained in a new position ($\text{MIF}_{\Delta \text{position}}$). $\text{MIF}_{\text{resting}}$ was calculated as the average IF over the 2 seconds preceding the onset of the ramp, and $\text{MIF}_{\Delta \text{position}}$ was calculated over the last 2

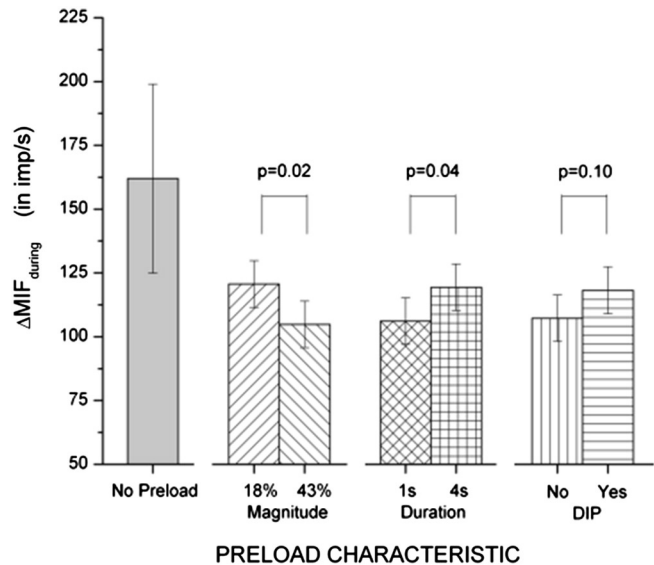


Fig 2. Effect of preload characteristics on muscle spindle responses during the manipulative thrust (mean and 95% CIs). Preload magnitudes represent percentage of peak applied thrust force. DIP, downward incisural point; $\text{MIF}_{\text{during}}$, mean instantaneous discharge frequencies during the manipulation.

seconds of the 4-second hold, which minimized mechanical contributions from the effects of tissue creep and spindle receptor adaptation to this metric.⁴⁰ Dynamic aspects were represented by the average spindle afferent IF calculated over the course of the entire ramp ($\text{MIF}_{\text{average movement}}$) and by the maximum spindle afferent IF during the ramp ($\text{MIF}_{\text{peak movement}}$) calculated by taking the average of the 3 largest IFs during the last half of the ramp. The latter excluded any initial burst arising from acceleration.^{39,40} Changes in the responsiveness of muscle spindles due to the preload and consequent to the manipulation were determined by subtracting MIFs during the first ramp and hold from MIFs during the second ramp and hold yielding the following response variables: $\Delta\text{MIF}_{\text{resting}}$, $\Delta\text{MIF}_{\Delta \text{position}}$, $\Delta\text{MIF}_{\text{average movement}}$, and $\Delta\text{MIF}_{\text{peak movement}}$.

Neural responses from the no preload conditions with and without HVLA-SM were compared with paired *t* tests. The response for the no preload condition with HVLA-SM was analyzed with a 1-sample *t* test. Neural responses were compared across the random presentations of the 8 types of preload combinations using a 1-way analysis of variance for the randomized complete block design with terms for block, main effects, 2-way interactions, and 3-way interaction (SAS System for Windows, version 9.2; SAS Institute, Inc, Cary, NC). Cat served as a blocking factor to control for the level of spindle activity and intra-animal variability. An overall F test was used to test whether the means were the same over the preload configurations. Overall F tests and post hoc contrasts using Dunnett's method to control for multiple comparisons when interactions were significant were tested at the 0.05 level of significance. Data are

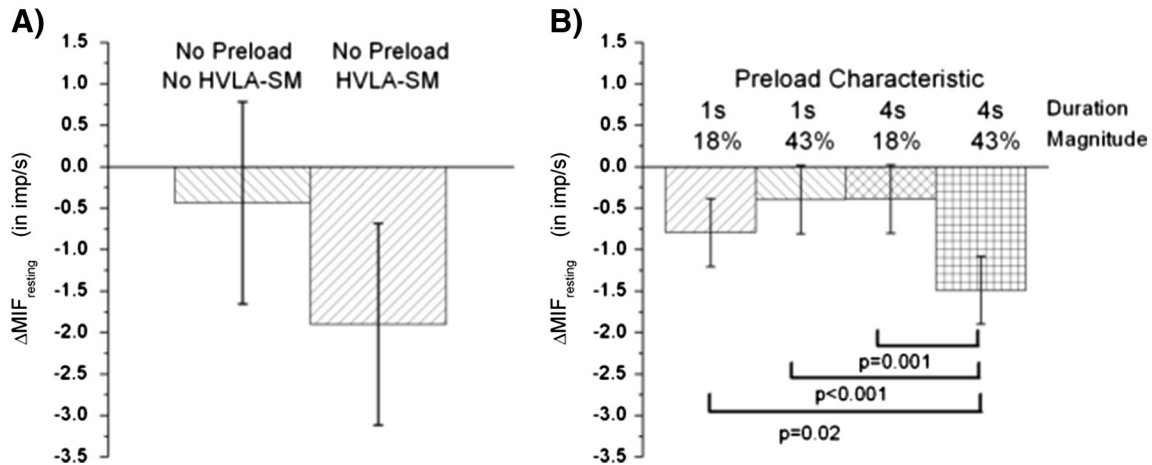


Fig 3. Effect of an HVLA-SM's preload characteristics on resting muscle spindle activity (mean and 95% CIs) measured before and after the manipulation. A, Control protocols for time (left) and preload (right). B, Changes following the HVLA-SM due to preload characteristics. HVLA-SM, HVLA-SM, high-velocity, low-amplitude spinal manipulation; $MIF_{resting}$, resting spindle discharge measured before the onset of the ramp.

reported as means and lower and upper 95% confidence intervals (LCI, UCI, respectively) based on the lower and upper analysis of variance model, unless otherwise noted.

RESULTS

Afferent recordings were obtained from 18 spindles in the lumbar longissimus muscle and 2 spindles in the multifidus muscle based upon criteria previously established.^{16-18,30,33} The most sensitive portion of each afferent's receptive field is shown in Figure 1. Most typically, this area was located at the level where the segmental muscles crossed the L₆₋₇ facet joint. All spindles entrained to the ~80-Hz vibratory stimuli. Nineteen spindles followed the stimulus with the vibrator pressed into the muscle belly through the overlying skin. One spindle responded at the first subharmonic (~40 Hz) through the skin but became entrained when the skin was removed and the vibrator pressed directly into the muscle belly. All afferents were silenced by muscle twitch using bipolar, direct muscle stimulation (amplitude, 0.2-0.3 mA; duration, 50 μ s). Following a single injection of succinylcholine (100 μ g/kg) into the carotid artery, the discharge of all but 1 afferent (which required 2 injections) increased by at least 40 imp/s within 8 seconds. The succinylcholine-induced discharge lasted at least 60 seconds. Consequently, the 20 afferents were classified as muscle spindle afferents.

Effect of Preload on Afferent Responses During an HVLA-SM (ΔMIF_{during})

There were no interactions among the 3 preload factors (magnitude, duration, and DIP) on muscle spindle responses during the manipulative thrust. Significant main effects occurred for preload magnitude ($F_{1,133} = 5.85$; $P = .02$) and preload duration ($F_{1,133} = 4.12$; $P = .04$). Shown in Figure 2,

the smaller preload magnitude significantly increased mean ΔMIF_{during} by 13.3 imp/s (0.3, 26.2, LCI, UCI), and the longer preload duration also significantly increased it by a mean of 15.8 imp/s (2.9, 28.7). Although not reaching statistical significance ($F_{1,133} = 2.78$; $P = .10$), the presence of a DIP, if anything, increased mean ΔMIF_{during} by 10.9 imp/s (-2.0, 23.9). Control protocols showed the absence of preload produced the greatest increase in spindle discharge during the manipulation with ΔMIF_{during} being at least 40 imp/s when compared with the presence of preload.

Effect of Preload on Static Responses ($\Delta MIF_{resting}$ and $\Delta MIF_{\Delta position}$) from Muscle Spindles Before and After an HVLA-SM

In the absence of an HVLA-SM (ie, due to time alone), average discharge for $\Delta MIF_{resting}$ spontaneously decreased by less than 1 imp/s, -0.44 (-1.65, 0.78) imp/s (Fig 3A). Following an HVLA-SM without preload, $\Delta MIF_{resting}$ decreased -1.90 imp/s (-3.29, -0.51). Paired t test showed that the difference between means (1.46 imp/s; -0.64, 3.57 [95% CI]) was not significant ($P = .16$). There was a significant interaction between the manipulation's preload magnitude and duration on $\Delta MIF_{resting}$ ($F_{1,133} = 10.85$; $P = .001$), but neither had a significant interaction with DIP ($F_{1,133} = 0.75$; $P = .39$ and $F_{1,133} = 0.06$; $P = .80$, respectively). Post hoc comparisons (Fig 3B) showed that the highest preload magnitude and longest duration led to a significantly greater mean decrease in resting spindle discharge than the other combinations. However, all decreases were modest in magnitude being similar to the control protocols (compare Fig 3B with 3A whose y-axis scales are the same). The main effect of DIP was not significant ($F_{1,133} = 2.11$, $P = .15$).

Mean average discharge for $\Delta MIF_{\Delta position}$ during the time control protocol (no HVLA-SM and no preload)

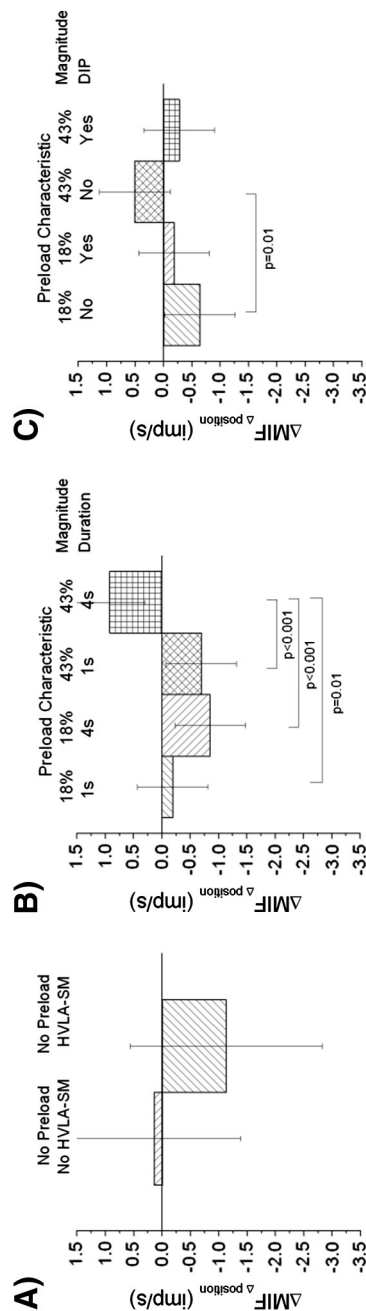


Fig 4. Effect of an HVLA-SM's preload characteristics on muscle spindle responses to a change in vertebral position (mean and 95% CIs) before and after the manipulation. A, Control protocols for time (left) and preload (right). B, Changes following the HVLA-SM due to interaction between preload magnitude and duration. C, Changes following the HVLA-SM due to interaction between preload magnitude and DIP. DIP, downward incisural point; HVLA-SM, high-velocity, low-amplitude spinal manipulation; MIF_{position} , spindle discharge measured during the hold when the vertebra was maintained in a new position.

spontaneously increased 0.14 (−1.38, 1.66) imp/s, whereas following the HVLA-SM without preload, it decreased, 1.12 (−0.56, 2.82) imp/s (Fig 4A). The difference between means was not significantly different from zero ($P = .20$). When HVLA-SM was delivered with preload, significant interactions between preload duration and preload magnitude ($F_{1,133} = 10.72$; $P = .001$) and between preload magnitude and preload trough ($F_{1,133} = 3.9$; $P = .05$) determined $\Delta MIF_{\text{position}}$. The highest preload magnitude (43% PATF) and longest duration (4 seconds) significantly increased spindle responsiveness to a new vertebral position when compared with all other combinations of preload magnitude and duration (Fig 4B). The highest preload magnitude (43% ABW) without a DIP compared with the lowest magnitude (18% PATF) without a DIP significantly increased spindle responsiveness to a new vertebral position (Fig 4C). $\Delta MIF_{\text{position}}$ appeared similar in the presence of a DIP regardless of preload magnitude (Fig 4C).

Effect of Preload on Dynamic Responses ($\Delta MIF_{\text{average movement}}$ and $\Delta MIF_{\text{peak movement}}$) From Muscle Spindles Before and After an HVLA-SM

Mean average discharge for $\Delta MIF_{\text{average movement}}$ during the time control protocol (no HVLA-SM and no preload) spontaneously increased 4.56 (2.04, 7.07) imp/s, whereas following the HVLA-SM without preload, it increased by only 2.41 (0.04, 4.42) imp/s (Fig 5A). The difference between means was significant ($P = .004$). When HVLA-SM was delivered with preload, there were no interactions. Significant main effects were present for preload magnitude ($F_{1,133} = 7.14$; $P = .01$) but not preload duration nor DIP, as shown in Fig 5B. The smaller preload magnitude (18% PATF) significantly increased mean $\Delta MIF_{\text{average movement}}$ more than the longer magnitude (43% PATF) by 0.60 imp/s (0.16, 1.05).

Similar to $\Delta MIF_{\text{average movement}}$ HVLA-SM without preload increased $\Delta MIF_{\text{peak movement}}$ less than the time control (1.72 [−1.14, 4.94] vs 6.17 [2.43, 9.91] imp/s) (Fig 6A). The difference between means was significant ($P = .009$). Preload duration, magnitude, nor DIP had a significant effect on $\Delta MIF_{\text{peak movement}}$ (Fig 6B).

DISCUSSION

Motivated by the idea that current scientific conclusions regarding SM's efficacy are limited by the paucity of preclinical studies that had informed those clinical trials of dosages to apply, we have undertaken a series of preclinical, basic science investigations using an animal preparation to identify the relationship between an SM's biomechanical characteristics and several neural^{16,33} and biomechanical^{20,41} responses that are currently thought to contribute to the mechanistic basis underlying the therapeutic benefits of SM. A goal of the current study is to motivate and provide a rational basis for either standardizing or identifying the

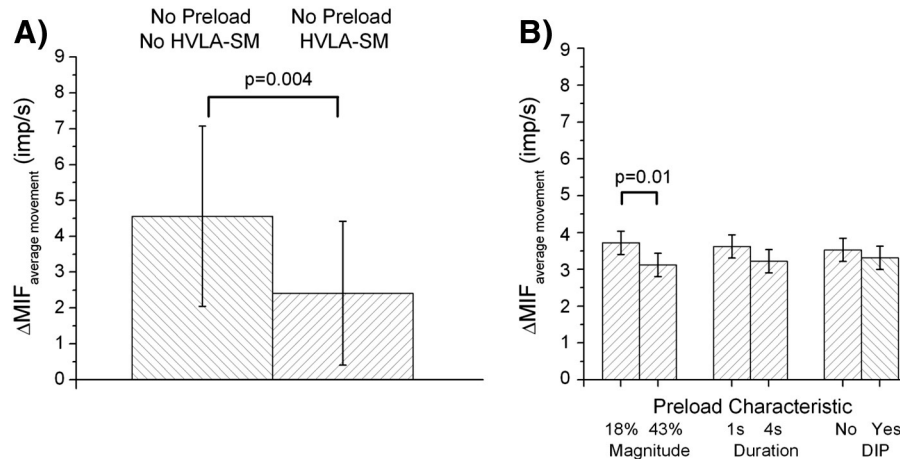


Fig 5. Effect of an HVLA-SM's preload characteristics on muscle spindle responses over the course of vertebral movement (mean and 95% CIs) before and after the manipulation. A, Control protocols for time (left) and preload (right). B, Changes following the HVLA-SM due to preload characteristics. DIP, downward incisural point; HVLA-SM, HVLA-SM, high-velocity, low-amplitude spinal manipulation; $MIF_{average movement}$, average spindle afferent instantaneous discharge frequencies calculated over the course of the entire ramp.

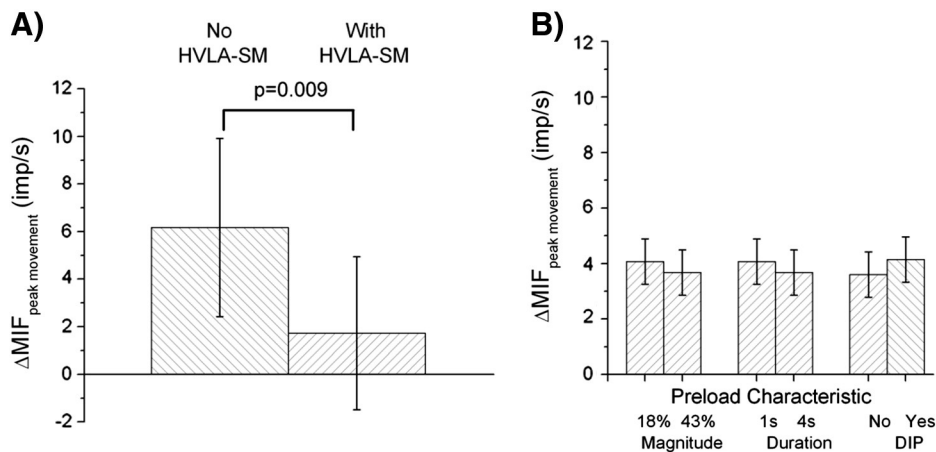


Fig 6. Effect of an HVLA-SM's preload characteristics on peak muscle spindle responses during vertebral movement (mean and 95% CIs) before and after the manipulation. A, Control protocols for time (left) and preload (right). B, Changes following the HVLA-SM due to preload characteristics. DIP, downward incisural point; HVLA-SM, HVLA-SM, high-velocity, low-amplitude spinal manipulation; $MIF_{peak movement}$, maximum spindle afferent instantaneous discharge frequencies during the ramp.

parameters that should be used to distinguish SMs delivered in clinical trials. Previous neurophysiological studies indicate that linear control of thrust displacement vs force as well as the SM's thrust duration, peak amplitude, and rate differentially affects the magnitude of sensory input from paraspinal muscle spindles during the SM¹⁶ but not their responsiveness following the SM.³³ Biomechanical studies suggest that thrust force and duration may interact to decrease spinal stiffness²⁰ and that the contact site at a given segmental level may modulate SM's effect on spinal stiffness.⁴¹ In the present study, the effect of preload characteristics on the magnitude of sensory input from paraspinal muscle spindles during and after the SM was systematically investigated.

Several preload characteristics affected spindle responses during the manipulative thrust. Our hypotheses

were based upon considering the preload as a form of conditioning that establishes muscle history. Muscle history has been consistently shown to alter the behavior of muscle spindles^{29,42} including those in paraspinal muscles.^{31,43} Lengthening histories stiffen the spindle apparatus by allowing stable, nonrecycling cross bridges to form between actin myosin filaments of the intrafusal fibers.²⁹ Further lengthening of a stiffened spine augments deformation of the spindle's equatorial region where the receptor potential is transduced and increases the responsiveness of the spindle.²⁹ Shortening following a history of lengthening creates slack and unloads the transducer region.

Our first hypothesis that increasing the preload magnitude would increase spindle responsiveness during the HVLA-SM was not supported (Fig 2). The smaller preload

magnitude unexpectedly evoked a significantly greater change in spindle discharge than the larger preload magnitude. The most likely explanation for this lies with the obligatory faster rate at which the manipulative thrust was applied following the lower preload. Following the lower preload magnitude, thrust force was delivered at 238 N/s because the applied thrust force rose 17.91 N (ie, PATF was fixed at 21.84 N, and the 18% PATF preloaded the tissues 3.93 N) was achieved in 75 milliseconds. Thrust rate during the 43% preload magnitude was given at 166 N/s because the applied thrust force rose only 12.45 N (21.84-9.39 N) in 75 milliseconds. An explanation based upon differences in thrust rate is also supported by the observation that the change in spindle discharge during the thrust when a preload was absent (gray bar in Fig 2) was greater than with either the 18% or 43% preload magnitudes. With the preload absent, thrust rate was 291 N/s ($21.84 \text{ N}/75 \text{ ms} = 291.2 \text{ N/s}$). Support for this explanation was also derived from a previous study³³ showing that muscle spindle discharge rapidly increases during an HVLA-SM when it is delivered at thrust rates between 50 and 250 N/s but tends to stabilize despite any additional increase in thrust rate (Fig 7B in ref³³). Muscle spindles in the lumbar paraspinal have been shown to be more sensitive to dynamic mechanical stimuli compared with appendicular muscles.⁴⁴ This high sensitivity appears to have dominated over the thixotropic behavior of the spindles apparatus.

Our second hypothesis that increasing preload duration would increase the discharge during the HVLA-SM was supported. The greater increase in $\Delta\text{MIF}_{\text{during}}$ with a preload duration of 4 seconds vs 1 second was consistent with a mechanism due to thixotropy. Unlike increasing preload magnitude, increasing preload duration did not decrease thrust rate because preload magnitude was constant during each of the 2 durations.

Our third hypothesis that the presence of a preload DIP will decrease muscle responsiveness during the HVLA-SM was not supported. The explanation for this likely lies with the temporal constraint that we imposed on thrust duration (75 milliseconds) and magnitude constraint that we imposed on peak thrust force (21.84 N), identical to what occurred when preload magnitude was reduced. The decrease in preload with DIP obligatorily led to an increase in thrust rate.

The values used in the current study for preload duration (1 and 4 seconds), preload magnitude (18% and 43% of peak applied thrust force), and preload DIP (9% of peak applied thrust force) can translate to clinical trials because they are based upon values used clinically. If 1 goal of manipulation is to normalize the behavior of central neurons by bombarding them with sensory input during the manipulation,⁸⁻¹¹ then preload duration appears to an important determinant of that sensory input from muscle spindles. Although preload magnitude and preload DIP may be important, the manner in which they experimentally had to be linked to thrust rate did not allow us to evaluate

their effects independent of changes in thrust rate. The results are consistent with previous observations that thrust rate is a defining determinant of muscle spindle discharge during a manipulative thrust.¹⁶

The effects of the preload characteristics on the static and dynamic signaling properties of muscle spindles following the HVLA-SM were not as dramatic as their effects during the HVLA-SM. Although each characteristic alone or through its interaction with the others often produced significant differences in signaling properties, the magnitudes of these differences were always small being typically less than 1 imp/s (Figs 3B, 4B, 5B, and 6B). It is difficult to know with any certainty whether such small differences are biologically important. It is worth noting that, as referred to above, muscle spindles in the lumbar paraspinal are more sensitive to both static and dynamic mechanical stimuli compared with appendicular muscles^{44,45} so that small changes in signaling may be proprioceptively or kinesthetically meaningful. The differences could warrant further investigation. However, spontaneous changes in muscle spindle signaling may account for some of these differences because the HVLA-SM in the absence of any preload did not significantly change any of the static or dynamic values compared with the time control protocol (Figs 3A, 4A, 5A, and 6A).

As noted in our previous studies,^{16,33} several aspects of our study may moderate the conclusions regarding physical parameters that contribute to defining the dosage of SM. First, SMs are typically delivered at a vertebra's end range of motion, distracting and increasing the separation between facet joint surfaces.^{5,12,46} We did not bring the manipulated vertebra (L_6) to its end range of motion because the magnitude of this movement would have stretched the nerve filament and torn it from the recording electrode. Second, findings in the present study were obtained with cats where the presence of a musculoskeletal condition or biomechanical condition for which SM might typically be applied clinically could not be established. Although clinically oriented basic science studies have shown that SM in asymptomatic human subjects induces physiologic responses, including reduced in H-reflexes⁴⁷ and increased pain thresholds,^{48,49} it remains possible that the absence of a spinal condition modified the effect on preload on muscle spindle responses to the HVLA-SM. It should also be recognized that this study was performed on passive muscle spindles where gamma-motoneuronal influences on the spindle were abolished. Korr⁵⁰ proposed that SM resets the gamma bias on muscle spindles, an effect that we would not have been able to see in the passive conditions established for this study. In addition, paraspinal tissues are innervated by low- and high-threshold mechanoreceptors other than muscle spindles, which likely respond to SM.⁵¹ Whether similar conclusions would arise from their responses needs additional investigation.

In conclusion, the preload that preceded an SM was characterized by its duration, magnitude, and whether it

contained a downward incisural dip. The effect that these characteristics had on sensory input from a spinal muscle proprioceptor was identified. A longer compared with a shorter preload duration produced a significantly larger increase in paraspinal muscle spindle discharge during the manipulative thrust. This effect likely arose from muscle history. A smaller compared with a larger preload magnitude produced a larger increase in paraspinal muscle spindle discharge; however, this response was opposite in direction to what would be expected from muscle history. This likely occurred from the faster manipulative thrust rate that accompanied the smaller preload magnitude. The presence of a preload DIP did not significantly affect muscle spindle discharge during the manipulative thrust. These data suggest that an SM's preload characteristics have consequences on the neural responses evoked during the manipulative thrust. On the other hand, the manipulation's preload characteristics had a small and likely inconsequential effect on producing sustained changes in the passive signaling properties of the muscle spindle following the manipulation. In clinical efficacy studies of HVLA-SM, consideration should be given to measuring and reporting preload characteristics and manipulative thrust rates because they appear to impact neural mechanisms thought to contribute to SM's therapeutic effects. In clinical practice, preload characteristics should be considered as an integral part of the intervention.

Practical Applications

- A longer compared with a shorter preload duration produced a significantly larger increase in paraspinal muscle spindle discharge during the manipulative thrust.
- A smaller compared with a larger preload magnitude produced a larger increase in paraspinal muscle spindle discharge.
- Preload characteristics should be taken into consideration when judging this intervention's therapeutic benefit in both clinical efficacy studies and in clinical practice.

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CONTRIBUTORSHIP INFORMATION

Concept development (provided idea for the research): JGP, GNR.

Design (planned the methods to generate the results): JGP, CRL.

Supervision (provided oversight, responsible for organization and implementation, writing of the manuscript): JGP.

Data collection/processing (responsible for experiments, patient management, organization, or reporting data): WRR.

Analysis/interpretation (responsible for statistical analysis, evaluation, and presentation of the results): CRL, JGP.

Literature search (performed the literature search): JGP, WRR.

Writing (responsible for writing a substantive part of the manuscript): JGP, WRR.

Critical review (revised manuscript for intellectual content, this does not relate to spelling and grammar checking): GNR, CRL.

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